

EXPERIMENT NO. 2 / 3

Data Flow Diagrams (Level 0, Level 1 & Level 2) & Data Dictionary

Project: Hospital Management System (HMS)

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1. Aim

To develop the Data Flow Diagrams (Context Level 0, Level 1, and Level 2) and prepare a comprehensive Data Dictionary for the Hospital Management System (HMS).

2. Objectives

- To understand the functional flow of healthcare and billing data across organizational boundaries.
- To model external entities, processes, data stores, and data flows using standard Gane-Sarson / DeMarco DFD notations.
- To decompose high-level processes into detailed functional sub-processes (Level 2 decomposition of Billing & Payment).
- To establish a comprehensive Data Dictionary cataloging all data stores, composite data flows, and atomic data elements.

3. DFD Notations Used

Notation Element	Geometric Symbol	Function in HMS
External Entity	Rectangle	Source or destination of data outside system boundary (e.g., Patient, Doctor, Payment Gateway).
Process	Circle / Rounded Rectangle	Transforms incoming data flows into outgoing data flows (e.g., 1.0 Patient Registration).
Data Store	Open-ended Rectangle / Database	Persistent repository holding healthcare records (e.g., D1 Patient Master, D6 Billing Store).
Data Flow	Directed Arrow	Pipeline carrying structured data packets between entities, processes, and data stores.

4. Level 0 DFD (Context Diagram)

The Context Diagram establishes the boundary of the Hospital Management System, portraying it as a single central process (0) communicating with six external entities: Patient, Receptionist, Doctor, Pharmacist & Lab Technician, Administrator, and the Payment Gateway.

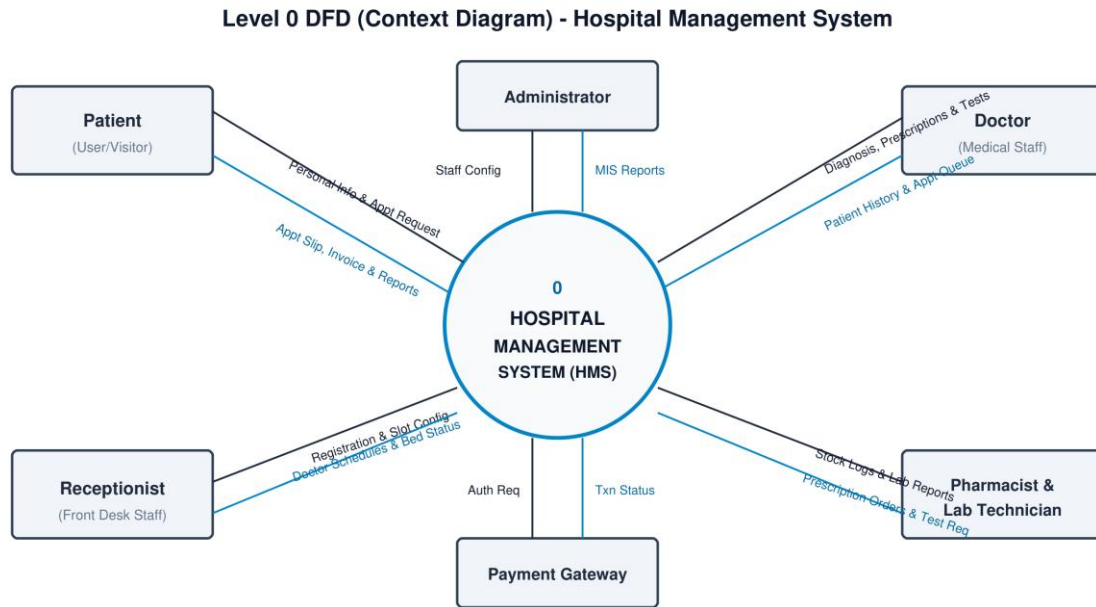


Fig 1: Level 0 DFD (Context Diagram) - Hospital Management System

5. Level 1 DFD (Decomposition into Functional Modules)

The Level 1 DFD decomposes the system into six core operational processes and seven centralized data stores:

- 1.0 Patient Registration & Appointment Management (Stores into D1 Patient Master & D2 Appointment Schedule)
- 2.0 Doctor Consultation & EHR Management (Reads D1, writes D3 Medical Records, triggers Lab & Pharmacy)
- 3.0 Laboratory & Diagnostic Management (Receives orders, writes to D4 Lab Reports Store)
- 4.0 Pharmacy & Stock Inventory Management (Reads/writes D5 Pharmacy Stock Store)
- 5.0 Billing & Payment Processing (Aggregates costs from D3, D4, D5, communicates with Payment Gateway, writes D6 Invoices)
- 6.0 Hospital Administration & MIS Reporting (Audit trail in D7, system monitoring)

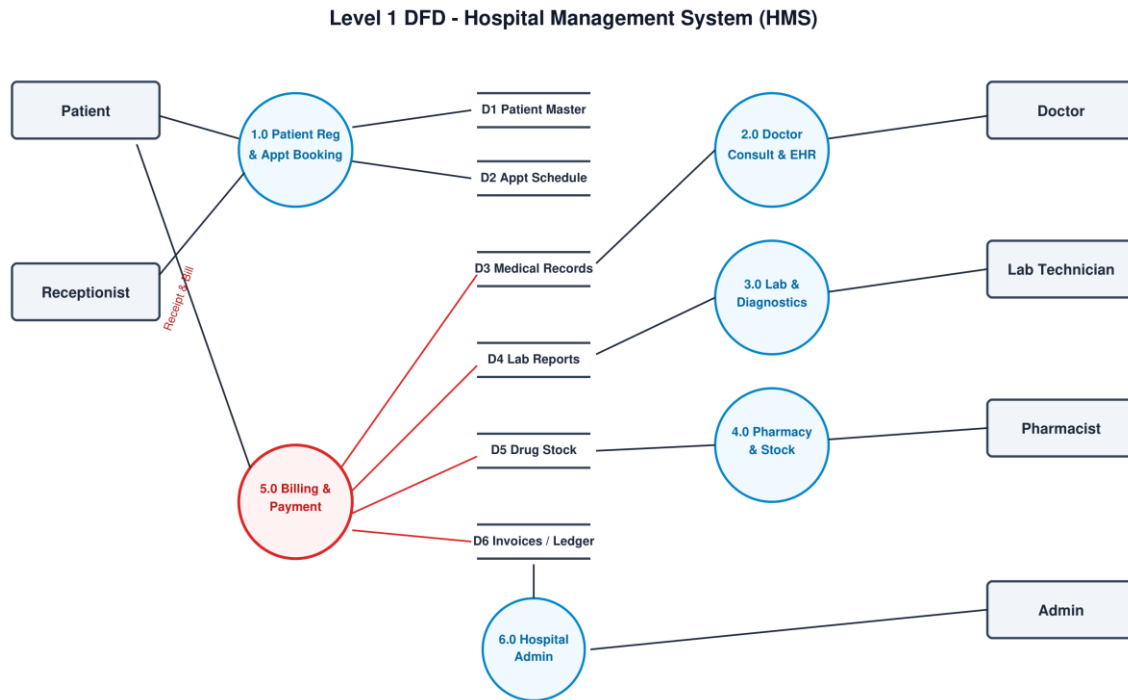


Fig 2: Level 1 DFD - Hospital Management System (Functional Decomposition)

6. Level 2 DFD (Decomposition of Process 5.0: Billing & Payment)

Process 5.0 is expanded into four sub-processes to reveal the inner mechanics of financial transactions:

- **5.1 Aggregate Service Charges:** Combines Doctor Consultation Fee (from D3), Lab Investigation Charges (from D4), and Dispensed Medicine Costs (from D5).
- **5.2 Verify Insurance & Apply Discounts:** Validates patient policy credentials and applies insurance co-pay discount rules.
- **5.3 Process Payment Transaction:** Handles cashier cash reception or transmits digital token to external Payment Gateway.
- **5.4 Generate Itemized Invoice & Update Ledger:** Updates accounting ledger D6 and generates digital receipt for Patient.

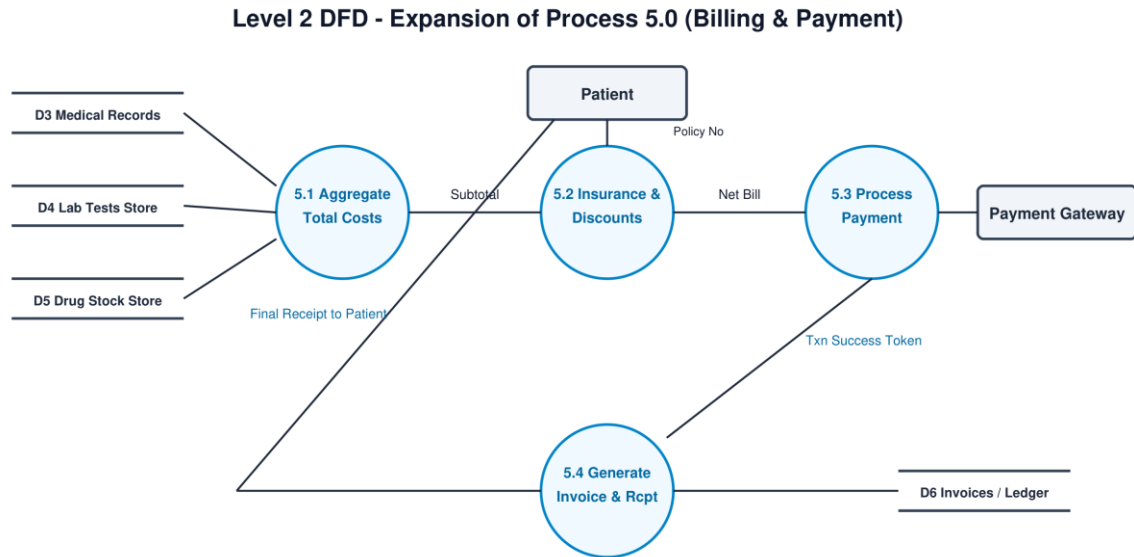


Fig 3: Level 2 DFD - Decomposition of Process 5.0 (Billing & Payment Processing)

7. Data Dictionary

7.1 Data Store Dictionary

Store ID	Data Store Name	Contents & Field Structure
D1	Patient Master Store	Patient_ID, Name, DOB, Gender, Contact_No, Address, Blood_Group, Emergency_Contact
D2	Appointment Schedule	Appointment_ID, Patient_ID, Doctor_ID, Appt_Date, Time_Slot, Booking_Status
D3	Medical Records & EHR	Record_ID, Patient_ID, Doctor_ID, Diagnosis, Symptoms, Prescription_ID, Consultation_Date
D4	Laboratory Reports	Test_ID, Patient_ID, Doctor_ID, Test_Name, Sample_Type, Test_Result, Status, Date
D5	Pharmacy Inventory	Medicine_ID, Brand_Name, Category, Stock_Quantity, Unit_Price, Batch_No, Expiry_Date
D6	Invoices & Ledger	Bill_ID, Patient_ID, Consultation_Fee, Lab_Cost, Med_Cost, Discount, Net_Payable, Txn_ID, Status
D7	Staff & Audit Log	User_ID, Employee_ID, Role, Password_Hash, Action_Logged,

Timestamp, IP_Address

7.2 Data Flow Dictionary (Composition)

Data Flow Name	Composition Formula (= composed of +)
Patient Registration Info	Patient_Name + DOB + Gender + Contact_No + Address + Blood_Group
Appointment Request	Patient_ID + Doctor_ID + Preferred_Date + Preferred_Time_Slot
Electronic Prescription	Prescription_ID + Patient_ID + Doctor_ID + {Medicine_Name + Dosage + Duration} + Instructions
Diagnostic Test Requisition	Test_ID + Patient_ID + Doctor_ID + Test_Name + Priority_Flag
Itemized Invoice & Receipt	Bill_ID + Patient_ID + Date + Fee + Lab_Cost + Med_Cost - Discount + Net_Amount + Txn-Token

8. Conclusion

The multi-level Data Flow Diagrams and Data Dictionary provide a rigorous structural blueprint of data movement throughout the Hospital Management System. The Level 2 expansion of billing confirms precise integration between clinical departments and automated accounting.